Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2 Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

C3 Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.

C4 Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

1 Community Festival Beverage Booth [C1-1 C2-2]

A community festival company must decide which beverage booth to operate before knowing how strong customer attendance will be. The company can choose either a basic booth or a premium booth, while festival attendance may turn out to be high or low. If the company chooses the basic booth, it expects to earn a profit of sixteen thousand US dollars when attendance is high and ten thousand US dollars when attendance is low. If it chooses the premium booth, it expects to earn a profit of twenty-nine thousand US dollars when attendance is high and five thousand US dollars when attendance is low.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C2 Construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1: Basic booth	A_2: Premium booth
States of nature	S_1: High attendance	S_2: Low attendance
Uncertain event	Actual festival attendance after the booth decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. Monetary measure: Profit in thousand US dollars for each alternative–state pair	

C2 - Payoff table [2]

Each supplied monetary payoff is placed at the intersection of its alternative and state:

$$P(A_1, S_1) = 16, \quad P(A_1, S_2) = 10,$$
$$P(A_2, S_1) = 29, \quad P(A_2, S_2) = 5.$$

Thus, in thousand US dollars, the completed payoff table is

Alternative	High S_1	Low S_2
Basic booth A_1	16	10
Premium booth A_2	29	5



2 Local Delivery Fleet [C1-1 C2-3]

A local delivery firm must choose between a bicycle fleet and an electric-van fleet before traffic conditions are known. Traffic may be clear or congested. The bicycle fleet earns 4 thousand CNY per completed delivery batch and is expected to complete 14 delivery batches in clear traffic and 10 in congested traffic. The electric-van fleet earns 6 thousand CNY

per completed delivery batch and is expected to complete 12 delivery batches in clear traffic and 8 in congested traffic. Clear traffic also adds a fixed scheduling bonus of 5 thousand CNY to either fleet, while congestion creates a fixed loss of 4 thousand CNY.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C2 Construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1**:** Bicycle fleet	A_2**:** Electric-van fleet
States of nature	S_1**:** Clear traffic	S_2**:** Congested traffic
Uncertain event	Actual traffic condition after the fleet decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand CNY for each alternative–state pair.	

C2 - Payoff table [3]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Bicycle fleet; A_2: Electric-van fleet	Alternative labels, in row order.
S_j	S_1: Clear traffic; S_2: Congested traffic	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Bicycle–clear: 14 batches; Bicycle–congested: 10 batches; Electric van–clear: 12 batches; Electric van–congested: 8 batches	Expected completed delivery batches for each fleet under each traffic condition.
$v^A(A_i)$	Bicycle fleet: 4 thousand CNY/batch; Electric-van fleet: 6 thousand CNY/batch	Earnings per completed delivery batch from the chosen fleet.
$f^S(S_j)$	Clear traffic: 5 thousand CNY; Congested traffic: -4 thousand CNY	Fixed scheduling bonus or loss associated with the traffic condition.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times v^A(A_1)$ $+ f^S(S_1)$ $= P_{ay}(A_1, S_1)$	$q(A_1, S_2) \times v^A(A_1)$ $+ f^S(S_2)$ $= P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1) \times v^A(A_2)$ $+ f^S(S_1)$ $= P_{ay}(A_2, S_1)$	$q(A_2, S_2) \times v^A(A_2)$ $+ f^S(S_2)$ $= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	14×4 $+5$ $=61$	10×4 -4 $=36$
A_2	12×6 $+5$ $=77$	8×6 -4 $=44$

Negative rates reduce the per-unit contribution; negative fixed amounts represent losses or commitments deducted once. A negative final payoff is a shortfall.

Final payoff table (profit in thousand CNY).

Alternative	Clear traffic (S_1)	Congested traffic (S_2)
Bicycle fleet (A_1)	61	36
Electric-van fleet (A_2)	77	44

3 Regional Food Delivery Service [C1-2 C2-5]

A regional food delivery company must choose between a standard delivery service and an express delivery service before customer demand is known. Demand may be high or low. The standard service earns 3 thousand GBP per completed delivery order and also provides a fixed operating contribution of 5 thousand GBP. The express service earns 5 thousand GBP per completed delivery order and provides a fixed operating contribution of 8 thousand GBP. High demand adds another 2 thousand GBP per completed order to either service, while low demand adds 1 thousand GBP per completed order. The standard service is expected to complete 15 orders under high demand and 9 orders under low demand; the express service is expected to complete 11 orders under high demand and 7 orders under low demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**

C2 Construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	A_1: Standard delivery service	A_2: Express delivery service
States of nature	S_1: High customer demand	S_2: Low customer demand
Uncertain event	Actual customer demand after the delivery service is selected	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand GBP for each alternative–state pair.	

C2 - Payoff table [5]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Standard delivery service; A_2: Express delivery service	Alternative labels, in row order.
S_j	S_1: High customer demand; S_2: Low customer demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Standard–high: 15 orders; Standard–low: 9 orders; Express–high: 11 orders; Express–low: 7 orders	Expected completed delivery orders for each service under each demand level.
$v^A(A_i)$	Standard service: 3 thousand GBP/order; Express service: 5 thousand GBP/order	Contribution per completed delivery order from the chosen service.
$v^S(S_j)$	High demand: 2 thousand GBP/order; Low demand: 1 thousand GBP/order	Additional contribution per completed order from customer demand.
$f^A(A_i)$	Standard service: 5 thousand GBP; Express service: 8 thousand GBP	Fixed operating contribution from the chosen service.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times$ $[v^A(A_1)+v^S(S_1)]$ $+f^A(A_1)$ $=P_{ay}(A_1, S_1)$	$q(A_1, S_2)\times$ $[v^A(A_1)+v^S(S_2)]$ $+f^A(A_1)$ $=P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1)\times$ $[v^A(A_2)+v^S(S_1)]$ $+f^A(A_2)$ $=P_{ay}(A_2, S_1)$	$q(A_2, S_2)\times$ $[v^A(A_2)+v^S(S_2)]$ $+f^A(A_2)$ $=P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$15\times$ $[3+2]$ $+5$ $=80$	$9\times$ $[3+1]$ $+5$ $=41$
A_2	$11\times$ $[5+2]$ $+8$ $=85$	$7\times$ $[5+1]$ $+8$ $=50$

Final payoff table (profit in thousand GBP).

Alternative	High customer demand (S_1)	Low customer demand (S_2)
Standard service (A_1)	80	41
Express service (A_2)	85	50

4 Local Caf'e Investment [C1-1 C3-2]

A small caf'e must decide whether to launch an artisan-dessert line or keep the current menu before foot traffic is known. Traffic may be high or low. All values are in million COP. Launching the dessert line produces revenue and cost of 58 and 16 under high traffic, and 27 and 19 under low traffic. Keeping the current menu produces revenue and cost of 49 and 21 under high traffic, and 41 and 21 under low traffic.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C3 Use revenue and costs to construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1: Launch dessert line	A_2: Keep current menu
States of nature	S_1: High traffic	S_2: Low traffic
Uncertain event	Actual foot traffic after the menu decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in million COP for each alternative–state pair.	

C3 - Calculating profit and building the payoff table [2]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Launch dessert line; A_2: Keep current menu	Alternative labels, in row order.
S_j	S_1: High traffic; S_2: Low traffic	State labels, in column order; revealed after the choice.
$R(A_i, S_j)$	Dessert–high: 58 million COP; Dessert–low: 27 million COP; Current menu–high: 49 million COP; Current menu–low: 41 million COP	Total revenue generated by each menu decision under each traffic condition.
$C(A_i, S_j)$	Dessert–high: 16 million COP; Dessert–low: 19 million COP; Current menu–high: 21 million COP; Current menu–low: 21 million COP	Total cost associated with each menu decision under each traffic condition.

Profit equation.

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$58-16$ $=42$	$27-19$ $=8$
A_2	$49-21$ $=28$	$41-21$ $=20$

Final payoff table (profit in million COP).

Alternative	High traffic (S_1)	Low traffic (S_2)
Launch dessert line (A_1)	42	8
Keep current menu (A_2)	28	20

5 Custom Furniture Product Line [C1-1 C3-3]

A custom-furniture workshop must choose between producing dining tables and producing office desks before market conditions are known. Conditions may be strong demand with favourable material prices or weak demand with higher material prices. The respective quantities are 7 and 4 production lots for dining tables, and 6 and 3 lots for office desks. Revenue per lot depends on the product: 14 thousand EUR for dining tables and 16 thousand EUR for office desks. Cost per lot depends on the state: 6 thousand EUR under strong demand and favourable prices, and 9 thousand EUR under weak demand and higher prices.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C3 Use revenue and costs to construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description
Alternatives	A_1: Produce dining tables A_2: Produce office desks
States of nature	S_1: Strong demand and favourable material prices S_2: Weak demand and higher material prices
Uncertain event	Market conditions revealed after the production decision
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand EUR for each product–market-state pair.

C3 - Profit and payoff table [3]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Produce dining tables; A_2: Produce office desks	Alternative labels, in row order.
S_j	S_1: Strong demand and favourable material prices; S_2: Weak demand and higher material prices	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Tables–strong: 7; Tables–weak: 4; Desks–strong: 6; Desks–weak: 3	Production lots for each product under each market condition.
$r(A_i)$	Dining tables: 14 thousand EUR/lot; Office desks: 16 thousand EUR/lot	Revenue per production lot.
$c(S_j)$	Strong demand: 6 thousand EUR/lot; Weak demand: 9 thousand EUR/lot	Cost per production lot.

Profit equation.

$$R = q(A_i, S_j)r(A_i), \qquad C = q(A_i, S_j)c(S_j), \qquad P(A_i, S_j) = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times r(A_1) = R(A_1, S_1)$	$q(A_1, S_2) \times r(A_1) = R(A_1, S_2)$
A_2	$q(A_2, S_1) \times r(A_2) = R(A_2, S_1)$	$q(A_2, S_2) \times r(A_2) = R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$7 \times 14 = 98$	$4 \times 14 = 56$
A_2	$6 \times 16 = 96$	$3 \times 16 = 48$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times c(S_1) = C(A_1, S_1)$	$q(A_1, S_2) \times c(S_2) = C(A_1, S_2)$
A_2	$q(A_2, S_1) \times c(S_1) = C(A_2, S_1)$	$q(A_2, S_2) \times c(S_2) = C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$7 \times 6 = 42$	$4 \times 9 = 36$
A_2	$6 \times 6 = 36$	$3 \times 9 = 27$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1) = P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2) = P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1) = P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2) = P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$98 - 42 = 56$	$56 - 36 = 20$
A_2	$96 - 36 = 60$	$48 - 27 = 21$

Final payoff table (profit in thousand EUR).

Alternative	Strong demand, favourable prices (S_1)	Weak demand, higher prices (S_2)
Produce dining tables (A_1)	56	20
Produce office desks (A_2)	60	21

6 Farm Subscription Plans [C1-2 C3-5]

A farm cooperative must choose between a standard produce box and a premium produce box before seasonal demand is known. Demand may be strong or weak. Quantities, in hundreds of subscriptions, are 8 and 5 for the standard box and 6 and 4 for the premium box under strong and weak demand, respectively. The plan-based revenue rates are 9 and 12 thousand USD per hundred subscriptions for the standard and premium boxes. Strong demand adds a state-based revenue rate of 3 thousand USD per hundred subscriptions; weak demand adds 1. The plan-based cost rates are 4 and 7 thousand USD per hundred subscriptions, respectively. The fixed distribution costs are 10 thousand USD under strong demand and 14 thousand USD under weak demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**

C3 Use revenue and costs to construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	A_1: Offer a standard produce box	A_2: Offer a premium produce box
States of nature	S_1: Strong seasonal demand	S_2: Weak seasonal demand
Uncertain event	Seasonal market conditions revealed after the subscription-plan decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand USD for each plan–market-state pair.	

C3 - Profit and payoff table

[5]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Offer a standard produce box; A_2: Offer a premium produce box	Alternative labels, in row order.
S_j	S_1: Strong seasonal demand; S_2: Weak seasonal demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Standard–strong: 8; Standard–weak: 5; Premium–strong: 6; Premium–weak: 4 hundreds of subscriptions	Subscription lots for each plan under each demand level.
$r^A(A_i)$	Standard box: 9 thousand USD/100 subscriptions; Premium box: 12 thousand USD/100 subscriptions	Plan-based revenue rate per hundred subscriptions.
$r^S(S_j)$	Strong demand: 3 thousand USD/100 subscriptions; Weak demand: 1 thousand USD/100 subscriptions	Additional revenue rate per hundred subscriptions from seasonal demand.
$c^A(A_i)$	Standard box: 4 thousand USD/100 subscriptions; Premium box: 7 thousand USD/100 subscriptions	Plan-based cost rate per hundred subscriptions.
$f_c(S_j)$	Strong demand: 10 thousand USD; Weak demand: 14 thousand USD	Fixed distribution cost under each demand condition.

Profit equation.

$$R = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right], \qquad C = q(A_i, S_j)c^A(A_i) + f_c(S_j), \qquad P(A_i, S_j) = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times [r^A(A_1)+r^S(S_1)] = R(A_1, S_1)$	$q(A_1, S_2)\times [r^A(A_1)+r^S(S_2)] = R(A_1, S_2)$
A_2	$q(A_2, S_1)\times [r^A(A_2)+r^S(S_1)] = R(A_2, S_1)$	$q(A_2, S_2)\times [r^A(A_2)+r^S(S_2)] = R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times [9+3] = 96$	$5\times [9+1] = 50$
A_2	$6\times [12+3] = 90$	$4\times [12+1] = 52$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times c^A(A_1) + f_c(S_1) = C(A_1, S_1)$	$q(A_1, S_2)\times c^A(A_1) + f_c(S_2) = C(A_1, S_2)$
A_2	$q(A_2, S_1)\times c^A(A_2) + f_c(S_1) = C(A_2, S_1)$	$q(A_2, S_2)\times c^A(A_2) + f_c(S_2) = C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times 4 + 10 = 42$	$5\times 4 + 14 = 34$
A_2	$6\times 7 + 10 = 52$	$4\times 7 + 14 = 42$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$96-42$ $=54$	$50-34$ $=16$
A_2	$90-52$ $=38$	$52-42$ $=10$

Final payoff table (profit in thousand USD).

Alternative	Strong seasonal demand (S_1)	Weak seasonal demand (S_2)
Standard produce box (A_1)	54	16
Premium produce box (A_2)	38	10

7 New Device Launch Strategy [C4-2]

A technology company must choose a strategy for launching a new wearable device. Market demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Demand		
Alternative	High	Medium	Low
Premium launch	90	38	-25
Standard launch	65	45	20
Economy launch	42	34	27
Delay launch	18	18	18

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin [2]



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_3$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	90	65	42	18	$A^*_{\text{Maximax}} = A_1$
Min	-25	20	27	18	$A^*_{\text{Maximin}} = A_3$

Maximax

$$\begin{aligned} M_1 &= \max\{90, 38, -25\} = 90 \\ M_2 &= \max\{65, 45, 20\} = 65 \\ M_3 &= \max\{42, 34, 27\} = 42 \\ M_4 &= \max\{18, 18, 18\} = 18 \end{aligned}$$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Premium launch)}$$

Maximin

$$\begin{aligned} m_1 &= \min\{90, 38, -25\} = -25 \\ m_2 &= \min\{65, 45, 20\} = 20 \\ m_3 &= \min\{42, 34, 27\} = 27 \\ m_4 &= \min\{18, 18, 18\} = 18 \end{aligned}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_3 \quad \text{(Economy launch)}.$$

8 Restaurant Expansion Strategy [C4-2]

A restaurant chain must choose a strategy for expanding its operations. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Customer demand		
Alternative	High	Medium	Low
Large expansion	85	32	-18
Moderate expansion	62	44	16
Small expansion	45	36	24
No expansion	20	20	20

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**

Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_3$

Maximax

$$\begin{aligned} M_1 &= \max\{85, 32, -18\} = 85 \\ M_2 &= \max\{62, 44, 16\} = 62 \\ M_3 &= \max\{45, 36, 24\} = 45 \\ M_4 &= \max\{20, 20, 20\} = 20 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{85, 32, -18\} = -18 \\ m_2 &= \min\{62, 44, 16\} = 16 \\ m_3 &= \min\{45, 36, 24\} = 24 \\ m_4 &= \min\{20, 20, 20\} = 20 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	85	62	45	20	$A^*_{\text{Maximax}} = A_1$
Min	-18	16	24	20	$A^*_{\text{Maximin}} = A_3$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Large expansion)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_3 \quad \text{(Small expansion).}$$

9 Renewable Energy Investment Strategy **[C4-2]**

A renewable-energy company must choose a strategy for developing a new solar project. Market conditions could be very favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

	Market conditions			
Alternative	Very favourable	Favourable	Stable	Unfavourable
Large solar farm	110	55	10	-35
Medium solar farm	82	60	28	5
Small solar farm	58	48	35	18
Partnership project	45	40	32	25

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_4$

Maximax

$$\begin{aligned} M_1 &= \max\{110, 55, 10, -35\} = 110 \\ M_2 &= \max\{82, 60, 28, 5\} = 82 \\ M_3 &= \max\{58, 48, 35, 18\} = 58 \\ M_4 &= \max\{45, 40, 32, 25\} = 45 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{110, 55, 10, -35\} = -35 \\ m_2 &= \min\{82, 60, 28, 5\} = 5 \\ m_3 &= \min\{58, 48, 35, 18\} = 18 \\ m_4 &= \min\{45, 40, 32, 25\} = 25 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	110	82	58	45	$A^*_{\text{Maximax}} = A_1$
Min	-35	5	18	25	$A^*_{\text{Maximin}} = A_4$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Large solar farm)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_4 \quad \text{(Partnership project).}$$